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TradePilot Root-Cause Analysis

Why the complex engine underperforms the simple original – full mechanism, change history, and the real fix direction

Project	TradePilot (paper-trading, NSE intraday)
Report	Root-Cause Analysis – engine performance regression
Version	v1.0.0
Status	Complete
Created	2026-06-24
Method	4 parallel read-only investigations (data path, trade lifecycle, git archaeology, loss correlation)
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1. Executive Summary

The investigation answers four questions: where stocks come from, how trades are targeted, what the original profitable engine did, and why current performance lags it.

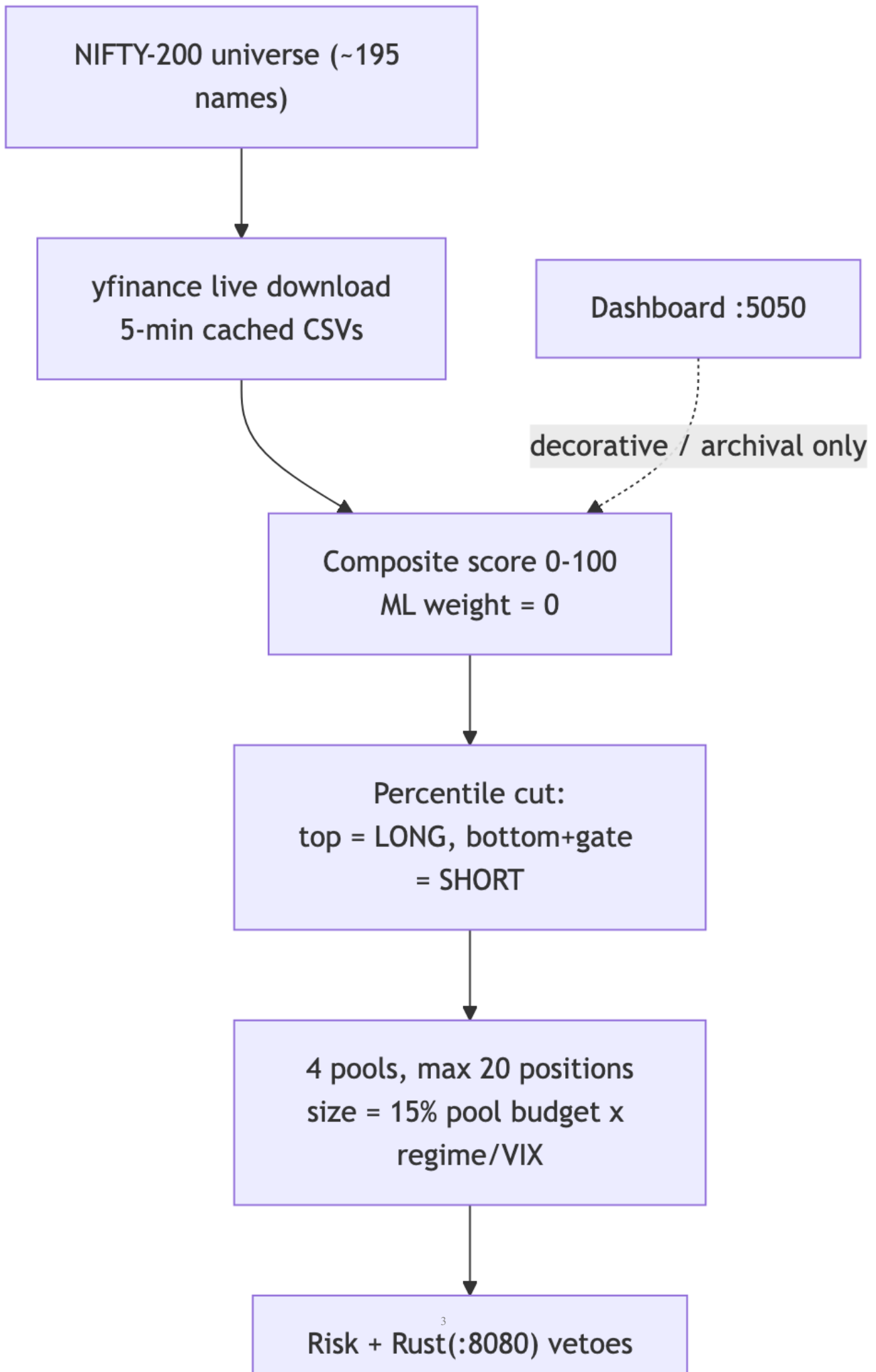
Three findings matter most:

1. **Stock data comes from live Yahoo Finance, not the dashboard.** The `localhost:5050` dashboard is decorative and archival – it is never in the live trade path.
2. **The original edge was selectivity and discipline, never the models.** On the best early day (82% win rate), winner and loser scores were statistically identical – the ML score was not picking winners. The tight bracket, long-only concentration, and end-of-day flat exit made the money.
3. **The complexity added since then diluted that edge.** Win rate fell from 82% to ~48% as trades-per-day rose from 17 to 45. That is the signature of over-diversification and over-trading regressing a selective alpha toward a coin flip.

A premise correction: the engine is **not currently losing money** – it is net positive over the last 12 sessions. The real problem is structural underperformance versus a far simpler version, and running at a fraction of target.

2. How Stock Selection Works Today

The live [v5](#) engine scores the NIFTY-200 (~195 names) off live Yahoo Finance downloads, ranks by a composite score, and opens positions through risk vetoes. The dashboard is not consulted.



Data and selection facts

ELEMENT	VALUE	SOURCE
Universe	NIFTY-200 (~195 names)	prototype/v4/config.py:97-99
Data feed	Live yfinance, 5-min cache	prototype/v4/data_nse.py:576
Dashboard in trade path	No – archival only	prototype/app.py (separate process)
Score model	Composite 0-100, ML weight 0	prototype/v4/config.py:166-167
Max positions	20 across 4 pools	prototype/v5/risk_manager.py:50

3. How Trades Are Targeted

Stops and targets are fixed-percent, set at signal time. Entries are spread across the day (initial deploy, then rescore every 30 minutes).

MECHANIC	VALUE	SOURCE
Stop-loss	-1.5% (-2.25% on gap mornings)	scripts/v5-paper-trade.py:491-493
Target	+2.0%	scripts/v5-paper-trade.py:494
Trailing	break-even at +1.0%, then 0.5% trail	scripts/v5-paper-trade.py:642-658
Position size	15% of pool budget x regime/VIX mult	scripts/v5-paper-trade.py:482
Rescore interval	30 min	scripts/v5-paper-trade.py:46

Exit reasons (evaluated in this order)

EXIT	TRIGGER	WHEN IT FIRES	RISK
FLAT_FORCE_EXIT	13:30-14:00 and abs(pnl) < 0.3%	Before stop/target are checked	Dumps recoverable near-flat positions at small loss + cost
STOPLOSS	Price crosses stop	Any scan	~1.5-2.25% realized loss
TARGET	Price reaches target	Any scan	Marks symbol re-armable
SIGNAL_FLIP	30-min rescore flips direction	After 60-min min-hold	Whipsaw if rescore is noisy
TIME_EXIT	15:15 IST	End of day	Intraday positions force-closed

4. What the Original Engine Did

The first profitable engine ([scripts/paper-trade-engine.py @ 236d6e4](#) , April) was dramatically simpler. Its measured result on 2026-04-08 was net +0.72% at an 82% win rate over 17 trades.

DIMENSION	ORIGINAL (APRIL)	CURRENT (JUNE)	DIRECTION
Universe	NIFTY-50	NIFTY-200	4x wider
Direction	Long-only	Long + short + flip	More exposure
Positions	5	20	4x more
Trades/day	17	~45	2.6x more
Bracket	+1.5% / -0.75%	+2.0% / -1.5%	Wider stops
Win rate	82%	44-53%	Collapsed

The complexity cascade was reactive: regime detection, kill switches, shorts, pools, the flip machine, and FLAT_FORCE_EXIT were each bolted on after a single bad day – most notably v4's -Rs 30,816 loss on 2026-04-09 (a bear day with no regime filter). None were added because the simple engine was failing.

5. The Root Cause

The original edge was **selectivity plus discipline, not intelligence**. Evidence: on the 82%-win-rate day, the average winner score (66.9) was effectively equal to the average loser score (69.2). The model was not separating winners from losers – the disciplined bracket and end-of-day flat exit on a concentrated long-only book did the work. This is corroborated twice: the ML information coefficient was 0.006, and ML is now weighted zero with no measured degradation.

Every complexity layer diluted that edge. As the engine widened to NIFTY-200, added shorts, grew to 45 trades/day, and spread capital across 20 positions in 4 pools, the win rate fell from 82% to ~48%. Rising trade count with falling win rate is the mathematical signature of over-diversification and over-trading: more bets on more marginal names, each paying ~12 bps round-trip cost, regressing the edge toward random.

The recurring SIDEWAYS-regime losses are a **symptom of the short book**, not a separate defect. The original was long-only and structurally could not lose on choppy days the way the current engine does – it had no shorts to short into the chop. Every recent red day was SIDEWAYS regime.

6. Premise Correction – The Engine Is Net Positive

The reported “5 consecutive losing days” is not supported by the result files. Over the last 12 sessions the live **v5** engine is net **+Rs 12,100 (+1.2% on Rs 10L)**. Red days are 4 of 12, scattered (not consecutive), and small (each <= 0.13% of capital). The two largest days are wins.

RED DAY	V5 (LIVE)	V5_CLASSIC (FROZEN ORIGINAL)	CAUSE
06-16	-1,347	-1,979 (worse)	Market / regime
06-17	-767	+2,088 (green)	v5-specific
06-18	-331	-1,605 (worse)	Market / regime
06-22	-1,050	+274 (green)	v5-specific

On two of the four red days the frozen original lost more than the current engine – so the added safeguards are net-ambiguous (they hurt on 06-17 and 06-22, helped on 06-16 and 06-18). The real issue is not bleeding; it is **structural underperformance versus the simple original**, and running at a fraction of the Rs 40-50k/day target (Rs 1.6k on a good day).

7. Recommended Fix Direction

This is an architecture question, not a patch. The pattern – each fix revealing a new problem elsewhere – indicates the reactive complexity cascade should be questioned rather than extended. All steps below are reversible and measured against real money-equivalent shadows.

ID	ACTION	PRIORITY
RC-1	Resurrect the original simple engine (paper-trade-engine.py @ 236d6e4) as a live shadow; run head-to-head vs v5 for 2 weeks	High
RC-2	Gate off shorts in SIDEWAYS regime (one-line change)	High
RC-3	Cut churn: drop position cap from 20 toward 5-8, tighten entry score threshold	Medium
RC-4	A/B FLAT_FORCE_EXIT off; pull v5_apr's 2-week record before deciding	Medium

The single highest-value experiment is **RC-1**: it requires no rebuild (the blueprint is in git), risks nothing (paper shadow), and directly tests the core hypothesis – that the simple, concentrated, long-only engine beats the complex one on win rate and on profit per rupee at risk.

8. Key Learnings

THEME	LEARNING	APPLICATION
Edge source	The original edge was execution discipline (tight bracket + flat exit + concentration), not the ML score	Do not add model complexity to chase alpha that lives in execution
Over-trading	Win rate 82% to 48% as trades 17 to 45 = over-diversification regressing alpha to coin-flip	Fewer, higher-conviction trades; treat trade count as a cost, not a virtue
Reactive complexity	Each safeguard was bolted on after one bad day; net effect is ambiguous	Resist single-incident-driven feature additions; A/B every safeguard
Shorts and regime	SIDEWAYS losses are a symptom of shorting into chop, absent in the long-only original	Regime-gate shorts before adding more short logic
Premise discipline	Owner's "5 losing days" was not in the data; engine is net +1.2%	Verify the symptom against result files before debugging the cause